



PLEDGE

I hereby pledge that I shall answer the questions at the examination to the best of my ability and that I shall not resort to any unfair means of any nature.

Invigilator's Signature : Student's Signature : Priya J. Shah

Course : Discrete Mathematics Name : Priya J. Shah

Date : 22/11/25 Student Identity Number : 202512100

Number of Supplementaries : 1 Programme : MScIT

Questions	1	2	3	4	5	6	7	8	9	10	TOTAL
Marks Obtained	0	5	20	15	00	20					
Questions	11	12	13	14	15	16	17	18	19	20	
Marks Obtained											60

Examiner's Signature

Start here

1. $M = \{1, \dots, 2n\}$, $n = \text{positive integer}$

show that $A \subset M$ cardinality $|A| = n+1$

has 2 distinct elements x and y

such that $x \mid y$

Let's prove it with an example

$$n = 2$$

$$2n = 4$$

So, $M = \{1, 2, 3, 4\}$

$$|A| = 2 + 1 = 3$$

All subsets with cardinality ≥ 3 can be A .

$$A_1 = \{1, 2, 3\}$$

$$A_3 = \{1, 3, 4\}$$

$$A_2 = \{1, 2, 4\}$$

$$A_4 = \{2, 3, 4\}$$

$$A_1 : x=1, y=3$$

$$1|3 : \text{Yes}$$

$$A_2 : x=2, y=4$$

$$2|4 : \text{Yes}$$

$$A_3 : x=1, y=3$$

$$1|3 : \text{Yes}$$

$$A_4 : x=2, y=4$$

$$2|4 : \text{Yes}$$

Therefore all subsets (A_1 to A_4) has 2 distinct elements x and y , such that x is dividing y .

$$\text{So } M = \{1, \dots, 2n\}$$

$$\text{with } A \subset M \text{ where } |A| = n+1$$

A have 2 distinct elements x and y , s.t. x divides y .
Hence, proved!

20

Total balls = 30

10 - Red

10 - Blue

10 - Green

$$\left(10 \times \frac{1}{2}\right) \times \left(20 \times \frac{1}{8}\right)$$

$$(5) \times (2.5)$$

$${}^{30}C_{10} \times \frac{1}{2} + {}^{30}C_{10} \times \frac{1}{8} + {}^{30}C_{10} \times \frac{1}{8}$$

$$\left(10 \times \frac{1}{2}\right)$$

$$\frac{10}{30} \left(\frac{1}{2}\right) + \frac{10}{30} \left(\frac{1}{8}\right) + \frac{10}{30} \left(\frac{1}{8}\right)$$

$$= \frac{1}{6} + \frac{1}{24} + \frac{1}{24}$$

5

$$= \frac{4 + 1 + 1}{24}$$

$$= \frac{6}{24} = 0.25$$

$\therefore P(A) =$ after 30 trials of picking a ball and putting it back in the bag, all the balls in bag are of same colour $\approx$ 0.25.

$$\frac{10}{60} = 0.167 \approx 0.17$$

$$\frac{17}{60} = 0.283 \approx 0.28$$

$$\frac{25}{60} = 0.416 \approx 0.42$$

$$\frac{12}{60} = 0.20$$

$$\frac{18}{60} = 0.30$$

$$\frac{30}{60} = 0.50$$

$$\frac{15}{60} = 0.25$$

$$\frac{20}{60} = 0.33$$

$$\frac{33}{60} = 0.55$$

Doubly stochastic 3×3 matrix

	C_1	C_2	C_3
R_1	$\frac{10}{60}$ (0.17)	$\frac{17}{60}$ (0.28)	$\frac{33}{60}$ (0.55)
R_2	$\frac{20}{60}$ (0.33)	$\frac{25}{60}$ (0.42)	$\frac{15}{60}$ (0.25)
R_3	$\frac{30}{60}$ (0.5)	$\frac{18}{60}$ (0.3)	$\frac{12}{60}$ (0.2)

$$R_1 = 0.17 + 0.28 + 0.55 = 1$$

$$R_2 = 0.33 + 0.42 + 0.25 = 1$$

$$R_3 = 0.50 + 0.30 + 0.20 = 1$$

$$C_1 = 0.17 + 0.33 + 0.50 = 1$$

$$C_2 = 0.28 + 0.42 + 0.30 = 1$$

$$C_3 = 0.55 + 0.25 + 0.20 = 1$$

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So the matrix is having 3 rows and 3 columns of non-negative entries

Sum of each row = Sum of each column = 1

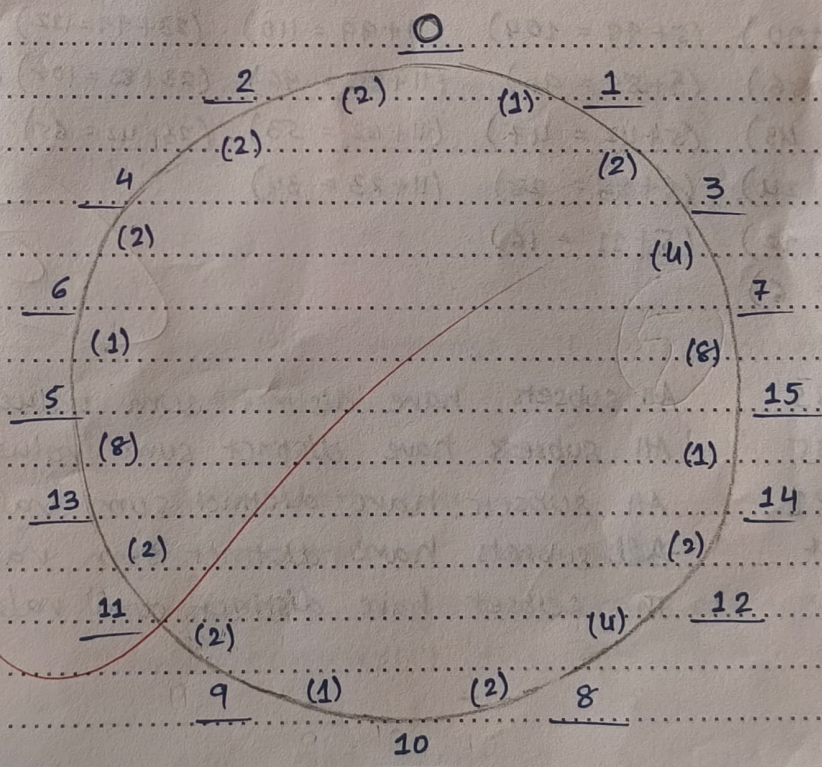
40

 $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$

Differ by : 1, 2, 4, 8 (any one)

Adjacent elements in cyclic order

→ Cyclic representation of numbers from 0 to 15 such that adjacent elements differ by (1, 2, 4, 8)



15

5. $|S| = 7$ where S is a set of positive integers

$$S = \{1, 5, 11, 23, 42, 85, 99\}$$

$$7C_1 = 7$$

All subsets have different ^{sum} values (distinct)

$$(1), (5), (11), (23), (42), (85), (99)$$

$$7C_2 = 21$$

All subsets have different ^{sum} values (distinct)

$$(1+99=100) \quad (5+99=104) \quad (11+99=110) \quad (23+99=122) \quad (42+99=141) \quad (85+99=184)$$

$$(1+85=86) \quad (5+85=90) \quad (11+85=96) \quad (23+85=108) \quad (42+85=127)$$

$$(1+42=43) \quad (5+42=47) \quad (11+42=53) \quad (23+42=65)$$

$$(1+23=24) \quad (5+23=28) \quad (11+23=34)$$

$$(1+11=12) \quad (5+11=16)$$

$$(1+5=6)$$

(14)

$$7C_3 = 35 \quad \text{All subsets have distinct sum values}$$

$$7C_4 = 35 \quad \text{All subsets have distinct sum values}$$

$$7C_5 = 21 \quad \text{All subsets have distinct sum values}$$

$$7C_6 = 7 \quad \text{All subsets have distinct sum values}$$

$$7C_7 = 1 \quad \text{The subset have distinct sum value}$$

6. S : set of non-empty integers

such that $T \subseteq S$

and the sum of elements in T is an odd number

→ So, the cardinality of S

$$|S| \in 1$$

- It will be a singleton set consisting of an odd element

- Therefore, $T \subseteq S$ will also have the same element
so $T = S$

- It will be singleton because if we increase the number of elements

case 1: Add even element

$$(\text{odd} + \text{even}) = \text{even odd}$$

but the singleton even element set will contradict the situation

case 2: Add odd element

$$(\text{odd} + \text{odd}) = \text{even}$$

$T \{a, b\}$ will be having even sum which will also contradict the situation

Hence, For eq:

$$S_1 = \{1, 2\}$$

$$T_1 = \{\emptyset\} \checkmark$$

$$T_2 = \{1\} \checkmark \quad T_3 = \{2\} \times$$

$$T_4 = \{1+2\} = \{3\} \checkmark$$

Due to T_3 the condition goes wrong

$$S_2 = \{1\}$$

$$T_1 = \{\emptyset\}$$

$$T_2 = \{1\}$$

Here, being a singleton set the condition is being managed.

Hence, $|S|=1$ must be there
for (every non-empty subset $T \subseteq S$, the sum of
elements in T is an odd number)
to be true.

20